

● EASY

● ADVANCED MATH

● ANSWER KEY

Advanced Math

06

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**A****A.** 3,100

Choice A is correct. The solution to a system of equations is the ordered pair x, y that satisfies all equations in the system. It's given by the first equation in the system that $x + 7 = 10$.

Substituting 10 for $x + 7$ into the second equation yields $10^2 = y$, or $y = 100$. The x -coordinate of the solution to the system of equations can be found by subtracting 7 from both sides of the equation $x + 7 = 10$, which yields $x = 3$. Therefore, the ordered pair 3, 100 is a solution to the given system of equations.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2**D****D.** $16w^2 - 2616w + 26$

Choice D is correct. The given expression follows the difference of two squares pattern, $x^2 - y^2$, which factors as $x - yx + y$. Therefore, the expression $256w^2 - 676$ can be written as $16w^2 - 26^2$, or $16w^2 - 26^2$, which factors as $16w - 2616w + 26$.

Choice A is incorrect. This expression is equivalent to $256w^2 - 832w + 676$.

Choice B is incorrect. This expression is equivalent to $64w^2 - 169$.

Choice C is incorrect. This expression is equivalent to $64w^2 - 208w + 169$.

Q3**3**

The correct answer is 3. A y -intercept of a graph in the xy -plane is a point x, y on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is 3. Therefore, the y -coordinate of the y -intercept of the graph shown is 3.

Q4**9**

The correct answer is 9. Dividing both sides of the given equation by 5 yields $|x| = 9$. By the definition of absolute value, if $|x| = 9$, then $x = 9$ or $x = -9$. Therefore, the two solutions to the equation $5x = 45$ are 9 and -9. It follows that the positive solution to the given equation is 9.

Q5**C**

C. $7x^3$

Choice C is correct. The given expression shows subtraction of two like terms. The two terms can be subtracted as follows: $12x^3 - 5x^3 = 12 - 5x^3$, or $7x^3$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the result of adding, not subtracting, the two like terms.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**C**

C. $Mt = 8902^{\frac{t}{10}}$

Choice C is correct. It's given that t represents the number of years since the account was opened. Therefore, $\frac{t}{10}$ represents the number of 10-year periods since the account was opened. Since the value of the account doubles during each of these 10-year periods, the value of the account can be found by multiplying the initial value by $\frac{t}{10}$ factors of 2. This is equivalent to $2^{\frac{t}{10}}$. It's given that the initial value of the account is \$ 890. Therefore, the value of the account Mt , in dollars, t years after the account was opened can be represented by $Mt = 8902^{\frac{t}{10}}$.

Choice A is incorrect. This equation represents the value of an account if the value of the account halves, not doubles, every 10 years.

Choice B is incorrect. This equation represents the value of an account if the value of the account decreases by 90%, not doubles, every 2, not 10, years.

Choice D is incorrect. This equation represents the value of an account if the value of the account increases by a factor of 10, not doubles, every 2, not 10, years.

Q7**117**

The correct answer is 117. Squaring both sides of the given equation gives $x+4=11^2$, or $x+4=121$. Subtracting 4 from both sides of this equation gives $x=117$.

Q8**D**

D. $15x^3 + 5x^2 + 5x + 2$

Choice D is correct. The given expression can be rewritten as $9x^3 + 6x^3 + 5x^2 + 5x + 7 - 5$. Combining like terms in this expression yields $15x^3 + 5x^2 + 5x + 2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q9**D****D.** 22

Choice D is correct. The value of $g4$ is the value of gx when $x = 4$. Substituting 4 for x in the given equation yields $g4 = 4 + 18$, which is equivalent to $g4 = 22$, or $g4 = 22$. Therefore, the value of $g4$ is 22.

Choice A is incorrect. This would be the value of $g4$ if function g was defined by $gx = -18$, not $gx = x + 18$.

Choice B is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x$, not $gx = x + 18$.

Choice C is incorrect. This would be the value of $g4$ if function g was defined by $gx = -x + 18$, not $gx = x + 18$.

Q10**C****C.** $r = 38s + t$

Choice C is correct. Multiplying both sides of the given equation by 38 yields $r = 38(s + t)$.

Choice A is incorrect. This equation is equivalent to $r - 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice B is incorrect. This equation is equivalent to $r + 38 = s + t$, not $\frac{r}{38} = s + t$.

Choice D is incorrect. This equation is equivalent to $38r = s + t$, not $\frac{r}{38} = s + t$.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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